

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Geochemistry Lab L-T-P: 0-0-3 Credits: 2

3. Pre-requisites: None

4. Syllabus and reference books:

Acid-base titration, Compleximetric EDTA titration for determining Ca, Mg, Fe; Preparation of rock samples for trace element measurements; acid-based digestion of rock powders; trace element measurements using ICP-MS; Ion exchange chromatographic separation of Sr and Nd, Sr-Nd isotope analyses using MC-ICPMS; In-situ trace element measurements in minerals using LA-ICP MS

Reference Books:

- 1) Using Geochemical Data: evaluation, presentation, interpretation by H. R. Rollinson
- 2) Isotopes-Principles and Applications by Gunter Faure and Teresa M. Mensing
- 3) A handbook of Silicate rock analysis by P.J. Potts

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Acid-base titration	1
2.	Complexometric EDTA titration for determining Ca, Mg, Fe	2
3.	Preparation of rock samples for trace element measurements	1
4.	Acid-based digestion of rock powders	1
5.	Trace element measurements using ICP-MS	2
6.	Ion exchange chromatographic separation of Sr and Nd	2
7.	Sr-Nd isotope analyses using MC-ICPMS	1
8.	In-situ trace element measurements in minerals using LA-ICP MS	2
Total number of hours		12*3 = 36